



**DELHI PUBLIC SCHOOL NEWTOWN**  
**SESSION 2023-24**  
**MONDAY TEST**

**CLASS : X**  
**SUBJECT: MATHEMATICS**

**FULL MARKS: 40**  
**DATE: 15.06.23**

**General Instructions:**

- The paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

**SECTION: A**

[ 5 x 1 = 5 ]

1 i) The value(s) of  $k$  for which the quadratic equation  $2x^2 - kx + k = 0$  has equal roots is (are)

- a) 4                      b) 0 only                      c) 8 only                      d) 0, 8

ii) The quadratic equation  $x^2 - 4x + 1 = 0$  has

- a) Equal roots    b) Rational roots    c) Irrational roots    d) Imaginary roots

iii) If  $x \in \mathbb{R}$ , the solution set of  $6 \leq -3(2x - 4) < 12$  is

- a)  $\{x : x \in \mathbb{R}, 0 < x \leq 1\}$     b)  $\{x : x \in \mathbb{R}, 0 \leq x < 1\}$   
c)  $\{0, 1\}$     d)  $\{1\}$

iv) If Rakesh opened a recurring deposit account in a bank and deposited ₹800 per month for 1 year 6 months, then the total money deposited in the account is

- a) ₹11400    b) ₹14400    c) ₹13680    d) ₹14100

v) If the rate of CGST is 9% , then the rate of GST will be

- a) 9%    b) 4.5%    c) 18%    d) 27%

**SECTION: B**

2. The marked price of an article is ₹18000 and the rate of GST on it is 12%. A shopkeeper buys this article at a reduced price and sells it at its marked price. If the shopkeeper paid ₹198.24 as CGST to the government, find the amount (inclusive of tax) paid by the shopkeeper . [4]

3. Mr Rakesh gets ₹6455 at the end of one year at the rate of 14% per annum in a Recurring Deposit Account. Find the monthly instalment. [4]

4. Solve the following equation for  $x$  and give your answer correct to 2 decimal places;  
 $x^2 + 6x - 10 = 0$  [4]

5. i) Simplify;  $\frac{-1}{3} \leq \frac{x}{2} - \frac{4}{3} < \frac{1}{6} : x \in \mathbb{R}$ . Graph the value of x on the real number line.  
 ii) Find the largest value of x for which  $2(x - 1) \leq 9 - x$  and  $x \in \mathbb{W}$ . [3 + 3]
6. Find the value of m ; if the roots of the following quadratic equations are equal :  
 $(4 + m)x^2 + (m + 1)x + 1 = 0$  [4]
7. A train travels a distance of 300 km at a constant speed. If the speed of the train is increased by 5km an hour, the journey would have taken 2 hours less. Find the speed of the train. [4]
8. Raghav deposited ₹200 per month for 36 months in a bank's recurring deposit account. If the bank pays interest at the rate of 11% per annum, find the amount he gets at maturity. [4]
9. For the following transaction of goods.

| Item | Quantity | Rate per item(₹) | Discount | GST |
|------|----------|------------------|----------|-----|
| A    | 1        | 2500             | 30%      | 18% |
| B    | 2        | 2000             | 20%      | 12% |
| C    | 1        | 1500             | NIL      | NIL |

Calculate the :

i)Total taxable amount.

ii)Total bill amount.

[5]